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Assessment of antibacterial activity of different solvent extracts of medicinal plant: *Aegle marmelos*

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Abstract Every part of Aegle marmelos is known for its several medicinal properties. In this present study invitro antibacterial activity of aqueous, methanol, ethanol and acetone extracts from leaves were investigated and the extracts exhibit broad spectrum of antibacterial activity with the zone of clearance ranging from 6-17mm to Salmonella sp, *Staphylococcus sp*, *Shigella sp* and *Pseudomonas* sp. Extracts of acetone and ethanol shows high susceptible to *Salmonella* and *Shigella* sp. The extracted *Aegle marmelo* shave antioxidant and cytotoxic potential were investigated by determined by 1,1-diphenyl - 2- picylhydroxyl radical scavenging activity, Hydroxyl radical scavenging activity and scavenging of hydrogen peroxidase. Phytochemical screening was performed to determine the ability of an extracts which inhibit the growth of bacteria.

Key Words *Aegle marmelos*, antibacterial activity, leaf extracts, Zone of inhibition

Introduction

A. marmelos is a native plant of India. *A. marmelos* fit in to Rutaceae family and commonly recognized as wood apple. In India, *A. marmelos* is grown as a temple garden plant and the leaves are charity to pray Lord Shiva. Bael (*Aegle marmelos*) is one of the most significant tree species used in numerousaboriginal systems of medicine in India, China, Burma and Sri Lanka (Kirtikar et al., 1984) and has been used by the inhabitants of Indian subcontinent for over 5000 years. In recent years antimicrobial assets of Indian medicinal plants have been increased (Aswal et al., 1996; Ahmad et al., 1998). The best practice to intervention is of much importance, in particular India due to the endemic presence of infective gastro intestinal diseases which are the major causes of infant and adult mortality (Miranda et al., 1993). Antibiotic principles are the distributed widely among angiosperm plants and variety of components were gathered in plant parts for their constitutive antimicrobial activities (Vlietinck, and Lindsay, 1995). Each and every part of *Aegle marmelos* plant like fruits, stem, bark, and leaves exhibit medicinal property and is used for treating eye and skin infections. Most accumulatory parts of the plant leaf contain bioactive compounds which are produced as secondary metabolites (Cowan, 1999). Nearly 100 phytochemical components have been isolated from different parts of the plant, namely phenols, flavonoids, alkaloids, cardiac glycosides, saponins, terpenoids, steroids and tannins. These components possess both biological and pharmacological activity against chronic diseases such as cancer and cardiovascular and gastrointestinal disorders (Badam et al., 2002). The present study was, therefore, aimed at evaluating the antimicrobial and antioxidant activity of *Aegle marmelos* leaf extracts against some

pathogenic microbes. Antimicrobial activities are due to the secondary metabolites synthesized by the plants such as phenolic compounds.Limited information is available regarding antimicrobial activity of *Aegle marmelos* leaves. The present studies to investigate of antibacterial activity of extract from the leaves of *Aegle marmelos* against various bacterial species. Preliminary phytochemical studies of these extracts are also under-taken to find out bioactive compounds having antimicrobial activity.

Materials and methods

Collection of plant samples

The fresh leaves of Aegle marmelos were collected from cultivated farms and the open fields of Theni district, Tamilnadu, India. Fresh parts of the plants were recognized and authenticated proceeding for phytochemical analysis. The leaves were rinsed thrice with distilled water followed by double distilled water to remove the dirt and other contaminants then the leaves were shade dried for 35 days and finally crushed into a fine powder. The leaf powder was stored in a well closed container for further studies.

Preparation of extracts

The leaf powder (20 g) was mixed with 100ml of different solvents such as acetone, ethanol, and methanol and aqueous in conical flask and kept on a rotary shaker for 24 hours at 30°c.

Phytochemical screening

Preliminary phytochemical screening for possible presence of bioactive compounds such as Alkaloids, Flavonoids, Terpenoids, Tannins, Saponins, Cardiac Glycosides, Steroids, Phlobatannins, Anthraquinones, Reducing sugar, Phenols, Coumarins, Anthocyanin and Betacyanin by standard methods.

Alkaloids

The extract was filtrates and treated with Dragendroff's reagent (solution of Potassium Bismuth Iodide). The formation of red color precipitate indicates the presence of alkaloids.

Amino acids

2 ml extract and 2 ml of ninhydrin solution was added in boiled for few minutes, the formation of blue color precipitate indicates the presence of amino acid.

Carbohydrates

The extract was filtrates were treated with two drops of alcoholic alpha-naphthol solution in test tube. The formation of precipitate the violet ring at the junction indicates the presence of carbohydrates.

Flavonoids

To the aqueous filtrate 5ml of dilute ammonia solution was added, followed by concentrated H_2SO_4. A yellow colouration indicated the presence of flavonoids.

Terpenoids

To 200 mg plant material 2 ml of chloroform and 3ml of concentrated sulphuric acid (H_2SO_4) were carefully added. A reddish brown colouration signified the presence of terpenoids.

Tannins

200 mg of plant material was boiled in 10ml distilled water and few drops of $FeCl_3$ were added to the filtrate; a blue-black precipitate indicated the presence of Tannins.

Saponins

5 ml dis. H_2O was added to 200mg extract material. 0.5ml filtrate was diluted with 5ml with dis. H_2O and shaken vigorously for 2 minutes. The formation of stable foam which is indicates the presence of saponins.

Cardiac glycosides

2 ml filtrate was treated with 1ml glacial acetic acid containing few drops of $fecl_3$.Conc. H_2SO_4 was added to the above mixture giving green-blue color depicting the positive results for presence of cardiac glycosides.

Steroids

200 mg plant extract was added in 10 ml chloroform. Acetic anhydride was added in the ratio of 1:1 which results convert into the formation of precipitate blue-green ring the presence of steroids.

Anthraquinones

500 mg of dried plant leaves were boiled in 10% HCl for 5min and filtrate was allowed to cool. Equal volume of chloroform with few drops of 10% NH_3 was added to 2 ml filtrate. The formation of rose-pink color implies the presence of anthraquinones

Phenols

1 ml of the extract was mixed with 5 ml of distilled water and few drops of neutral Ferric chloride were added to this mixture. The appearance of dark green color indicated the presence of phenol.

Antioxidant Assay

Antioxidant potential of Aegle marmelos extract was determined by following methods.

DPPH (1, 1-diphenyl-2-picrylhydrazyl) radical scavenging activity

DPPH assay was measured by spectrophotometry. DPPH assay give reliable information concerning the antioxidant ability of the tested compound. Aliquots of 20-100 µl (5 mg/10 ml) was added to 2.9 ml of freshly prepared solution of DPPH (6×10-5M in MeOH). After incubation at 37◦ C for 1 hour the absorbance was recorded at λmax 517 nm using spectrometer.

Formula

Percentage of inhibition or Scavenging activity of DPPH = (1-B/A) %

Where, B = Absorbance was taken by control solution, A = Absorbance was taken by different concentration of solution.

Scavenging of Hydrogen peroxide (H_2O_2)

The assay was performed for hydrogen peroxide in fruit and root extracts of A. *Marmelos* was following standard procedures. The antioxidant capacity of fruit extract against H_2O_2 value was expressed as mole of ascorbate equivalent per kg fresh weight. 0.6 ml of 4 mm hydrogen per oxide (in phosphate buffer solution) was added to 4 ml solution of fruits & leaf of *A.* m*armelos* and incubated for 10 minute. The absorbances of the solutions were measured at λ_{max} 230 nm against a blank solution containing hydrogen per oxide without extract.

Isolation and identification of organism

The sample was collected from local Lab in Theni. By serial dilution technique the organisms were isolated and undergo for biochemical and morphological identification of organisms. The isolates were maintained in nutrient agar plates for further studies.

Antimicrobial activity

Antimicrobial activity was performed using the well-diffusion. The plant extracts were tested on Mullen Hinton agar plates to detect the presence of antibacterial activity.

Bacterial suspensions for each of the tested organism were prepared in 9ml sterile nutrient broth and were incubated at 37ºC for 18hrs to get a turbidity of 0.5 Macfarland standard procedure. The bacterial cell suspension was corresponding to the spread on the surface of MHA (Mullen Hinton agar) plates with sterile cotton swab and kept to dry. The antibacterial assay was done by agar diffusion technique, as a result of five equally distant 6mm wells were made on the inoculated MHA plates with the help of a sterile cork borer. Each well was loaded on 50µl of different solvent with extract respectively using micropipette, the extract was allowed to diffuse for 30mts at room temperature and the loaded plates were then incubated at 37ºC for 18 to 24hrs. Standard without plant extract was used as a control. The appearance of zone which is indicated to the presence of antimicrobial activity of the plant extracts being tested against the bacterial isolates.

Result and discussion

Phytochemical screening

The four extracts revealed the presence of different phytochemicals such as tannins, saponins, flavonoids, alkaloids, trepenoids, phenol, reducing sugars, amino acids and phylobatanninsand the results were tabulated in the table 1. The medicinal plants containing phenolic compounds with tannins as the major ingredients were used as for care and repair of skin wounds (Dweck, 2002). Tannins are considered as nutritional and unwanted, because the precipitate of proteins, digestive enzymes and disturb the absorption of vitamins and minerals. Furthermore, the benefits which include the absence of adverse effects and a low incidence of resistance (Spann et al., 2003)

Antioxidant activity

The antioxidant activity of Aegle marmelos might be due to the inactivation of free radicals on complex formation with the metal ions results the phytochemical screening of aqueous, acetone, methanol and ethanol revealed the availability of multipolar and no polar chemical components. DPPH assay is a stable free radical method. It is an easy, rapid and sensitive path to survey the antioxidant activity of plant extracts (Koleva et al., 2002). The reduction DPPH takes place due to the presence of hydrogen contribution as antioxidant by the formation diphenypicrylhydrozine. It reduces the color of DPPH with the help of power donating ability. Hydrogen peroxide is not a reactive and it sometimes cause the cytotoxicity and increases the hydroxyl inside the cell thus removes the hydrogen peroxide and scavenging the aqueous, ethanol, methanol and acetone extract. Antioxidant activity of leaf extract in various solvent was tabulated in the table 2.

Isolation and identification of organisms

The isolated pathogens were identified as *Salmonella*, *Shigella*, *Pseudomonas* and *Staphylococcus* sp by Bergey's manual classification of bacteria and the results were tabulated in the table 3.

Antibacterial activity

The extract from solvents such as ethanol, methanol, acetone showed unreliable antibacterial activity against the test organisms (Table 4a, 4b, 4c & 4d) Well diffusion method revealed maximum inhibition against the *Salmonella*, *Shigella*, *Pseudomonas* and

Staphylococcus sp. It is observed that methanol, ethanol extracts of leaf shows maximum inhibitory effect against *Shigella* and *Salmonella* sp. In the acetone extract *Staphylococcus* sp shows maximum susceptibility when compared to other organisms. Most antimicrobial medicinal plants are more active against Gram-positive than Gram-negative bacteria (Lin et al., 1999; Srinivasan et al., 2001).

Conclusion

The activities of aqueous, methanol, acetone and ethanol extracts of *Aegle marmelos* on bacterial pathogens such as *Salmonella*, *Shigella*, *Pseudomonas* and *Staphylococcus* sp.as discovered in this present study it can be used as traditional therapy.

Table 1: Phytochemical screening of extracts from leaves

Compounds	Methanol Extract	Ethanol Extract	Acetone Extract	Aqueous Extract
Alkaloids (Dragendroff's test)	A	A	A	P
Flavonoids (Ammonia Test)	A	P	P	P
Tannins (Ferric Chloride Method)	P	P	P	A
Terpenoids (Salkowski test)	P	A	A	P
Phenol (Ferric Chloride Method)	P	P	P	A
Steroid (Liebermann – Burchard reaction test)	P	P	P	A
CardiacGlycosides (Keller – Kiliami Test)	A	P	P	A
Anthraquinone (Borntrager's Test)	A	A	A	A
Carbohydrate (Molisch's test)	A	A	P	P
Saponins (Frothing Test)	P	A	P	P
Amino acids (Ninhydrin Method)	P	P	P	P

P=Presence, A=Absence

Table 2: Antioxidant activity

S. No	Concentration (µg/ml)	DPPH Scavenging Activity				H_2O_2 Radical assay			
		Met	Eth	Ace	Aqu	Met	Eth	Ace	Aqu
1	50	15.5	16.7	13.2	8.2	15	15.9	14.7	9.5
2	100	21	20.8	19.6	13	20.5	19.2	20.2	12.3
3	150	26.3	25.1	22.9	17.3	25	23.6	24.6	18.5
4	200	32.7	28.3	27.5	22.9	35.2	27.8	28.9	23.9
5	250	38	34	30.4	25.7	40.3	30.1	32.5	29.2

Met- Methanol, Eth- Ethanol, Ace- Acetone, Aqu- aqueous

Table 3: Identification of microbial pathogen

Characterization	*Salmonella sp*	*Shigella sp*	*Pseudomonas sp*	*Staphylococcus sp*
Morphological Characterization				
Simple Staining	Rod	Rod	Rod	Cocci
Gram Staining	Gram Negative	Gram Negative	Gram Negative	Gram positive
Spore Staining	Non sporing	Non sporing	Non sporing	Non sporing
Motility	Motile	Non motile	Motile	Non motile
Capsule Staining	Non -Capsulated	Non-Capsulated	Non -Capsulated	Non -capsulated
Biochemical Characterization				
Indole test	Negative	Positive	Negative	Negative
Urease test	Negative	Negative	Negative	Positive
Citrate	Negative	Negative	Positive	positive

Catalase	Negative	Negative	Positive	Positive
Oxidase	Negative	Negative	Positive	negative
VP Test	Negative	Negative	Negative	Positive
MR test	Positive	Positive	Negative	Positive
Cultural characterization				
Specific Media	Black colony on SS Agar	Pink colony on SS Agar	White colony on cetrimide Agar	Yellow Colony on Mannitol Salt Agar

Table 4(a): Antimicrobial activity of aqueous leaf extract

Concentration of extract	***Staphylococcus sp***	***Salmonella sp***	***Pseudomonas sp***	***Shigella sp***
	Zone of Inhibition in mm			
50µl	2	3	1	2
100µl	5	4	4	3
150µl	6	7	7	6
200µl	9	9	8	9
250µl	11	10	9	11

Table 4(b): Antimicrobial activity of methanol extract

Concentration of extract	***Staphylococcus sp***	***Salmonella sp***	***Pseudomonas sp***	***Shigella sp***
	Zone of Inhibition in mm			
50µl	9	10	10	9
100µl	11	13	10	12
150µl	14	12	12	14
200µl	15	17	13	16
250µl	17	19	16	18

Table 4(c): Antimicrobial activity of ethanol extract

Concentration of extract	***Staphylococcus sp***	***Salmonella sp***	***Pseudomonas sp***	***Shigella sp***
	Zone of Inhibition in mm			
50µl	9	9	8	9
100µl	11	10	12	13
150µl	13	12	13	14
200µl	14	13	14	15
250µl	15	17	16	17

Table 4(d): Antimicrobial activity of acetone extract

Concentration of extract	***Staphylococcus sp***	***Salmonella sp***	***Pseudomonas sp***	***Shigella sp***
	Zone of Inhibition in mm			
50µl	10	9	8	8
100µl	12	10	9	11
150µl	13	12	11	12
200µl	15	14	12	15
250µl	17	16	14	16

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